

Applets2\BotonesApplet.java

```
1  package Applets2;
2
3  import javax.swing.*;
4  import java.awt.event.*;
5  import java.awt.*;
6
7  public class BotonesApplet extends JFrame implements ActionListener {
8
9      private JTextField txt1, txt2, txtResult;
10     private JButton btnAdd, btnSubtract, btnMultiply, btnDivide, btnClear, sqRt1, sqRt2,
11     major, minor;
12
13     public BotonesApplet() {
14         setTitle("Calculadora Swing - Extender Resultado");
15         setSize(400, 600);
16         setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
17
18         setLayout(new BorderLayout(10, 10));
19
20         JPanel panelTop = new JPanel(new GridLayout(2, 2, 10, 5));
21         panelTop.setBorder(BorderFactory.createEmptyBorder(10, 10, 10, 10));
22
23         txt1 = new JTextField();
24         txt2 = new JTextField();
25         txt1.setFont(new Font("Arial", Font.BOLD, 20));
26         txt2.setFont(new Font("Arial", Font.BOLD, 20));
27
28         panelTop.add(new JLabel("Número 1:", SwingConstants.CENTER));
29         panelTop.add(new JLabel("Número 2:", SwingConstants.CENTER));
30         panelTop.add(txt1);
31         panelTop.add(txt2);
32
33         JPanel panelCentralContenedor = new JPanel(new BorderLayout(10, 10));
34         panelCentralContenedor.setBorder(BorderFactory.createEmptyBorder(0, 10, 10, 10));
35
36         // Sub-panel para los botones
37         JPanel panelButtonsGrid = new JPanel(new GridLayout(3, 3, 10, 10));
38
39         btnAdd = new JButton("Sumar");
40         btnSubtract = new JButton("Restar");
41         btnMultiply = new JButton("Multiplicar");
42         btnDivide = new JButton("Dividir");
43         sqRt1 = new JButton("Raiz de 1");
44         sqRt2 = new JButton("Raiz de 2");
45         major = new JButton("Mayor");
46         minor = new JButton("Menor");
47         btnClear = new JButton("Limpiar");
```

```
48 // Listeners
49 JButton[] botones = {btnAdd, btnSubtract, btnMultiply, btnDivide, sqRt1, sqRt2,
major, minor, btnClear};
50 for (JButton btn : botones) {
51     btn.addActionListener(this);
52     panelButtonsGrid.add(btn);
53 }
54
55 JPanel panelResultadoFila = new JPanel(new BorderLayout(10, 10));
56 JLabel lblRes = new JLabel("Resultado: ");
57 lblRes.setFont(new Font("Arial", Font.BOLD, 14));
58
59 txtResult = new JTextField();
60 txtResult.setEditable(false);
61 txtResult.setBackground(Color.LIGHT_GRAY);
62 txtResult.setFont(new Font("Arial", Font.BOLD, 20));
63 txtResult.setHorizontalAlignment(JTextField.LEFT);
64
65 panelResultadoFila.add(lblRes, BorderLayout.WEST);
66 panelResultadoFila.add(txtResult, BorderLayout.CENTER);
67
68 panelCentralContenedor.add(panelButtonsGrid, BorderLayout.CENTER);
69 panelCentralContenedor.add(panelResultadoFila, BorderLayout.SOUTH);
70
71 add(panelTop, BorderLayout.NORTH);
72 add(panelCentralContenedor, BorderLayout.CENTER);
73
74 setLocationRelativeTo(null);
75 setVisible(true);
76 }
77
78 @Override
79 public void actionPerformed(ActionEvent e) {
80     try {
81         if (e.getSource() == btnClear) {
82             txt1.setText("");
83             txt2.setText("");
84             txtResult.setText("");
85             return;
86         }
87
88         double num1 = Double.parseDouble(txt1.getText());
89         double num2 = 0;
90
91         if (!txt2.getText().isEmpty()) {
92             num2 = Double.parseDouble(txt2.getText());
93         }
94
95         double result = 0;
96
```

```
97         if (e.getSource() == btnAdd) {
98             result = num1 + num2;
99         } else if (e.getSource() == btnSubtract) {
100             result = num1 - num2;
101         } else if (e.getSource() == btnMultiply) {
102             result = num1 * num2;
103         } else if (e.getSource() == btnDivide) {
104             if (num2 != 0) {
105                 result = num1 / num2;
106             } else {
107                 JOptionPane.showMessageDialog(this, "No se puede dividir por cero",
108 "Error", JOptionPane.ERROR_MESSAGE);
109                 return;
110             }
111         } else if (e.getSource() == sqRt1) {
112             result = Math.sqrt(num1);
113         } else if (e.getSource() == sqRt2) {
114             result = Math.sqrt(num2);
115         } else if (e.getSource() == major) {
116             result = Math.max(num1, num2);
117         } else if (e.getSource() == minor) {
118             result = Math.min(num1, num2);
119         }
120
121         txtResult.setText(String.valueOf(result));
122     } catch (NumberFormatException ex) {
123         JOptionPane.showMessageDialog(this, "Por favor ingrese números válidos", "Error",
124 JOptionPane.ERROR_MESSAGE);
125     }
126
127     public static void main(String[] args) {
128         new BotonesApplet();
129     }
130 }
```